

PRACTICAL WEEK - 3

Task -1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
siddu@DESKTOP-5CRM05P: ~/practical3 X Ubuntu X + X
siddu@DESKTOP-5CRM05P:~$ mkdir practical3
mkdir: cannot create directory 'practical3': File exists
siddu@DESKTOP-5CRM05P:~$ cd practical3
siddu@DESKTOP-5CRM05P:~/practical3$ nano process.c
siddu@DESKTOP-5CRM05P:~/practical3$ gcc task1.c -o task1
cc1: fatal error: task1.c: No such file or directory
compilation terminated.
siddu@DESKTOP-5CRM05P:~/practical3$ gcc process.c -o process
siddu@DESKTOP-5CRM05P:~/practical3$ ./process
Before fork()
Current Process PID: 706

--- PARENT PROCESS ---
Parent PID: 706
Child PID: 707
Parent process is waiting...

--- CHILD PROCESS ---
Child PID: 707
Parent PID (PPID): 706
Child process is running
Child process has finished.
Parent process is ending.
```

```
GNU nano 8.7.1 process.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Before fork()\n");
    printf("Current Process PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        // Child process
        printf("\n--- CHILD PROCESS ---\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID (PPID): %d\n", getppid());
        printf("Child process is running\n");
    }
    else {
        // Parent process
        printf("\n--- PARENT PROCESS ---\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent process is waiting...\n");
        wait(NULL);

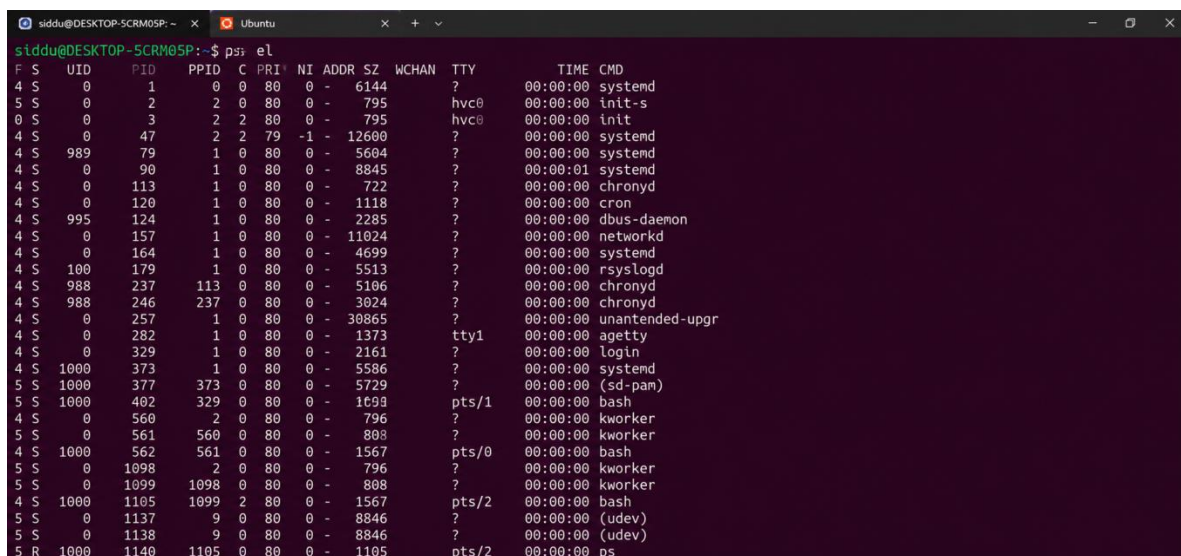
        printf("Child process has finished.\n");
        printf("Parent process is ending.\n");
    }
    return 0;
}
```

TASK - 2 : Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
siddu@DESKTOP-5CRM05P:~/practical3$ nano process.c
siddu@DESKTOP-5CRM05P:~/practical3$ gcc process.c -o process
siddu@DESKTOP-5CRM05P:~/practical3$ ./process
Before fork()
Current Process PID: 1092

--- PARENT PROCESS ---
Parent PID: 1092
Child PID: 1093
Parent process is waiting...

--- CHILD PROCESS ---
Child PID: 1093
Parent PID (PPID): 1092
Child process is running
Child process has finished.
Parent process is ending.
```



```
siddu@DESKTOP-5CRM05P:~$ ps -e
  F S      UID          PID    PPID    C PRI  NI ADDR SZ  WCHAN    TTY          TIME CMD
  4 S      0            1         0   0  80   0 -   6144      ?           00:00:00 systemd
  5 S      0            2         0   0  80   0 -   795      hvcd0       00:00:00 init-s
  0 S      0            3         0   0  80   0 -   795      hvcd0       00:00:00 init
  4 S      0           47         0   0  79  -1 - 12600      ?           00:00:00 systemd
  4 S    989           79         0   0  80   0 -   5604      ?           00:00:00 systemd
  4 S      0           90         0   0  80   0 -   8845      ?           00:00:01 systemd
  4 S      0          113         0   0  80   0 -   722      ?           00:00:00 chronyd
  4 S      0          120         0   0  80   0 -   1118      ?           00:00:00 cron
  4 S    995          124         0   0  80   0 -   2285      ?           00:00:00 dbus-daemon
  4 S      0          157         0   0  80   0 -  11024      ?           00:00:00 networkd
  4 S      0          164         0   0  80   0 -   4699      ?           00:00:00 systemd
  4 S    100          179         0   0  80   0 -   5513      ?           00:00:00 rsyslogd
  4 S    988          237        113   0  80   0 -   5106      ?           00:00:00 chronyd
  4 S    988          246        237   0  80   0 -   3024      ?           00:00:00 chronyd
  4 S      0          257         0   0  80   0 -  30865      ?           00:00:00 unattended-upgr
  4 S      0          282         0   0  80   0 -   1373    tty1       00:00:00 agetty
  4 S      0          329         0   0  80   0 -   2161      ?           00:00:00 login
  4 S   1000          373         0   0  80   0 -   5586      ?           00:00:00 systemd
  5 S   1000          377        373   0  80   0 -   5729      ?           00:00:00 (sd-pam)
  5 S   1000          402        329   0  80   0 -   1699    pts/1       00:00:00 bash
  4 S      0          560         0   0  80   0 -   796      ?           00:00:00 kworker
  5 S      0          561        560   0  80   0 -   808      ?           00:00:00 kworker
  4 S   1000          562        561   0  80   0 -   1567    pts/0       00:00:00 bash
  5 S      0         1098         0   0  80   0 -   796      ?           00:00:00 kworker
  5 S      0         1099        1098   0  80   0 -   808      ?           00:00:00 kworker
  4 S   1000         1105        1099   0  80   0 -   1567    pts/2       00:00:00 bash
  5 S      0         1137         0   0  80   0 -   8846      ?           00:00:00 (udev)
  5 S      0         1138         0   0  80   0 -   8846      ?           00:00:00 (udev)
  5 R   1000         1140        1105   0  80   0 -   1105    pts/2       00:00:00 ps
```

```
siddu@DESKTOP-5CRM05P: ~ X Ubuntu
siddu@DESKTOP-5CRM05P:~$ top
top - 05:52:31 up 29 min, 1 user, load average: 0.02, 0.01, 0.00
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7776.6 total, 7131.4 free, 559.2 used, 237.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7217.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24576   15540  11568 S   0.0   0.2   0:00.84 systemd
    2 root        20   0    3180    2204   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2040   1940 S   0.0   0.0   0:00.00 init
   47 root        19  -1   50400   16764   15468 S   0.0   0.2   0:00.23 systemd-journal
   79 systemd+    20   0   22416   14500   12012 S   0.0   0.2   0:00.10 systemd-resolve
   90 root        20   0   35380   12176   9080 S   0.0   0.2   0:01.40 systemd-udev
  113 root        20   0    2888    1948   1820 S   0.0   0.0   0:00.05 chronyd-starter
  120 root        20   0    4472    3176   2924 S   0.0   0.0   0:00.01 cron
  124 message+    20   0    8820    5404   4724 S   0.0   0.1   0:00.07 dbus-daemon
  157 root        20   0   44096   29800   14660 S   0.0   0.4   0:00.28 networkd-dispat
  164 root        20   0   18436    9428   8184 S   0.0   0.1   0:00.11 systemd-logind
  179 syslog      20   0  220548    5904   4612 S   0.0   0.1   0:00.07 rsyslogd
  237 _chrony      20   0   20424    6448   5668 S   0.0   0.1   0:00.07 chronyd
  246 _chrony      20   0   12096    2424   1780 S   0.0   0.0   0:00.00 chronyd
  257 root        20   0  123460   32604  17996 S   0.0   0.4   0:00.06 unattended-upgr
  282 root        20   0    5492    2756   2572 S   0.0   0.0   0:00.01 agetty
  329 root        20   0    8644    5612   4772 S   0.0   0.1   0:00.02 login
  373 shreyaj     20   0   22344   13604  10984 S   0.0   0.2   0:00.18 systemd
  377 shreyaj     20   0   22916    4076   2076 S   0.0   0.1   0:00.00 (sd-pam)
  402 shreyaj     20   0    6136    5420   3856 S   0.0   0.1   0:00.04 bash
  560 root        20   0    3184    1116    980 S   0.0   0.0   0:00.00 SessionLeader
  561 root        20   0    3200    1256   1108 S   0.0   0.0   0:00.06 Relay(562)
  562 shreyaj     20   0    6268    5540   3864 S   0.0   0.1   0:00.09 bash
 1098 root        20   0    3184    1116    980 S   0.0   0.0   0:00.00 SessionLeader
 1099 root        20   0    3200    1256   1108 S   0.0   0.0   0:00.02 Relay(1105)
 1105 shreyaj     20   0    6268    5612   3936 S   0.0   0.1   0:00.06 bash
 1156 shreyaj     20   0    8320    6108   3896 R   0.0   0.1   0:00.34 top
 1237 root        20   0   35384    6460   3332 S   0.0   0.1   0:00.00 (udev-worker)
 1238 root        20   0   35384    6460   3332 S   0.0   0.1   0:00.00 (udev-worker)
```